

Project Task Allocation & Workflow Matrix

Machine Learning Classification & GUI System • Team Responsibilities Plan

TEAM MEMBER ROLES & DELIVERABLES

Team Member	Domain & Role	Assigned Responsibilities & Deliverables
Abdussamad S	PROJECT LEAD & CORE ML / GUI	• End-to-end task allocation, tracking, and milestone verification • Decision Tree model development, hyperparameter tuning & evaluation • Comparative analysis against benchmark models • Tkinter GUI design, interactive workflow, and full system integration • Final team review, integration validation, and demonstration
Sanjana M	REQUIREMENTS & PREPROCESSING LEAD	• Facilitate team discussions to align specifications and scope • Review and validate data cleaning & feature preprocessing pipelines • Ensure transformed data meets modeling standards and schema constraints
Aditya K	DATA ENGINEERING & DOCUMENTATION	• Dataset tracking, versioning, and rigorous data quality validation • Coordinate data cleaning execution and Exploratory Data Analysis (EDA) • Author and maintain project technical documentation and system logs
Keerti C	REQUIREMENTS & EDA COORDINATION	• Core requirement gathering, functional spec definition, and tracking • Liaise between EDA outcomes, statistical findings, and ML developers • Ensure feature distributions and transformations align with ML prerequisites
Namrata D	KNN MODELINGPRESENTATION	• Coordinate and implement K-Nearest Neighbors (KNN) model development • Prepare baseline metrics, distance metric evaluations, and confusion matrices • Design presentation slide deck and compile comprehensive PDF deliverables

PHASE-WISE WORKFLOW & PIPELINE COORDINATION

Phase 1: Requirements Gathering & Dataset Setup

Keerti C and **Sanjana M** establish functional project scope and data needs. Concurrently, **Aditya K** acquires the source dataset, initiates version tracking, and sets baseline data quality benchmarks.

Phase 2: Data Cleaning, Preprocessing & EDA

Aditya K oversees missing-value imputation, outlier management, and EDA. **Sanjana M** verifies data pipelines for correctness. **Keerti C** channels EDA findings and prepared features directly to model engineers.

Phase 3: Machine Learning Model Development & Benchmarking

Abdussamad S develops the Decision Tree classifier while **Namrata D** coordinates KNN model tuning. **Abdussamad S** conducts comparative evaluation (Accuracy, F1-score, Precision, Recall, ROC-AUC) to select the optimal model.

Phase 4: Tkinter GUI Integration, Final Documentation & Demonstration

Abdussamad S implements the Tkinter GUI and integrates the trained model. **Aditya K** finalizes documentation, **Namrata D** delivers the slide deck and PDF export, and **Abdussamad S** verifies deliverables for the final team demo.